

ELEC5460 — Homework 1

Lectures 2–3: Probability Foundations and Fading Channels

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Due: (2026-10-08, 23:59)

Instructions. This set contains **eight long problems**; each has four connected parts (a)–(d) of increasing difficulty, and the final part of each question builds directly on the earlier ones. Work through the parts *in order*. Show *all* derivations in full rigor and state every assumption explicitly; partial credit is given only for correctly justified intermediate steps. Numerical answers must be accompanied by a closed-form expression. Computer algebra / numerical linear algebra (e.g. matrix inverses and DFTs) may be used in Q5, but every matrix or transform that is fed to software must be written out explicitly first. Question 2 and part (d) of Question 1 are conceptual-depth material marked $[\star]$ (*not examinable*; recommended for students continuing to research). All eight questions carry equal weight.

Notation and standing definitions. λ denotes Lebesgue measure on $\mathbb{R}$; $\mathcal{B}(\mathbb{R})$ are the Borel sets; $Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = \Phi(-x)$ is the Gaussian tail; for a complex Gaussian vector $z \sim \mathcal{CN}(0, \Sigma)$, $\mathbb{E}[zz^H] = \Sigma$ and $\mathbb{E}[z^H z] = \text{tr}(\Sigma)$. Symbols not defined in a problem carry their meaning from the lecture slides, but every *object the question asks you to derive from* is defined below in the problem itself.

Q1. Conditional laws from the bivariate Gaussian density

Setup. Let $Z_1, Z_2 \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ and define, for constants $\mu_X, \mu_Y \in \mathbb{R}$, $\sigma_X, \sigma_Y > 0$, $\rho \in [-1, 1]$,

$$X = \mu_X + \sigma_X Z_1, \quad Y = \mu_Y + \sigma_Y (\rho Z_1 + \sqrt{1 - \rho^2} Z_2).$$

Thus (X, Y) is a *bivariate Gaussian vector*: we say $(X, Y) \sim \mathcal{N}(\mu, \Sigma)$ with mean $\mu = (\mu_X, \mu_Y)^T$ and covariance $\Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}$. For part (a) you may use the multivariate change-of-variables formula: if $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a C^1 bijection with C^1 inverse, then the density of $(U, V) = T(X, Y)$ is $f_{U,V}(u, v) = f_{X,Y}(T^{-1}(u, v)) |\det DT^{-1}(u, v)|$. The *conditional density* of X given $Y = y$ (for y with $f_Y(y) > 0$) is defined by $f_{X|Y}(x|y) := f_{X,Y}(x, y)/f_Y(y)$, and the conditional mean/variance by $\mathbb{E}[X|Y = y] = \int x f_{X|Y}(x|y) dx$, $\text{Var}[X|Y = y] = \int (x - \mathbb{E}[X|Y = y])^2 f_{X|Y}(x|y) dx$.

- (a) Derive the joint density $f_{X,Y}(x, y)$ explicitly by writing (X, Y) as a linear map applied to (Z_1, Z_2) . Then verify $\iint f_{X,Y} dx dy = 1$ by integrating over x first, using the substitution $x = \mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y) + \sigma_X \sqrt{1 - \rho^2} t$ for each fixed y . (*This single substitution is the master trick for the whole question.*)
- (b) From the joint density of (a), compute $f_{X|Y}(x|y)$ by completing the square in x . Hence, find the conditional mean $\mathbb{E}[X|Y] = g(Y)$ and the conditional variance; verify $\mathbb{E}[g(Y)] = \mu_X$ (tower property; you may use $\mathbb{E}[g(Y)] = \int g(y) f_Y(y) dy$), and prove that $g(Y)$ is the minimum-MSE estimator of X from Y .
- (c) **The degenerate case.** Let $\rho = 1$ (the case $\rho = -1$ is identical up to sign). Show that $\det \Sigma = 0$ and prove that Y is an *affine function of X almost surely*. *Hint: Compute $\text{Var}(Y - \frac{\sigma_Y}{\sigma_X} X)$ and*

use that $\text{Var}(Z) = 0$ implies $Z = \mathbb{E}Z$ almost surely; then prove that $Y = \mu_Y + \frac{\sigma_Y}{\sigma_X}(X - \mu_X)$ a.s. Then show that (X, Y) has no density with respect to two-dimensional Lebesgue measure λ^2 , by exhibiting the witnessing set $L = \{(x, \mu_Y + \frac{\sigma_Y}{\sigma_X}(x - \mu_X)) : x \in \mathbb{R}\}$: $\lambda^2(L) = 0$ yet $\mathbb{P}((X, Y) \in L) = 1$. (Recall from Lecture 2: $\mu_{(X,Y)} \ll \lambda^2$ fails iff some λ^2 -null set carries positive mass.)

(d) [★] **How to write the “pdf” when Σ is rank-deficient.**

- (i) Define the one-dimensional reference measure $\nu := T_{\#}\lambda$ on L , the *push-forward* of Lebesgue measure under $T(t) = (t, \mu_Y + \frac{\sigma_Y}{\sigma_X}(t - \mu_X))$, i.e. $\nu(A) := \lambda(T^{-1}(A))$ for every Borel set $A \subseteq L$ (the image of λ under T); show from the definition of absolute continuity (“ $\nu(B) = 0 \Rightarrow \mu_{(X,Y)}(B) = 0$ ”) that $\mu_{(X,Y)} \ll \nu$, and show that the R-N density is $g = d\mu_{(X,Y)}/d\nu$, i.e. that

$$\mu_{(X,Y)}(B) = \int_B g d\nu, \quad g(t) = \frac{1}{\sigma_X \sqrt{2\pi}} e^{-(t - \mu_X)^2 / (2\sigma_X^2)}.$$

Verify this on two families of test sets: the one-parameter family $B_u = \{T(t) : t \leq u\} \cap L = \{T(t) : t \leq u\}$ (a tail-segment of the line), and rectangles $B = (a_1, b_1) \times (a_2, b_2)$ chosen so that $B \cap L$ is a single interval of L .

- (ii) Explain why the shorthand $f_{X,Y}(x, y) = \frac{1}{\sigma_X \sqrt{2\pi}} e^{-(x - \mu_X)^2 / (2\sigma_X^2)} \delta(y - \mu_Y - \frac{\sigma_Y}{\sigma_X}(x - \mu_X))$ is *not* a probability density (no such function exists in the sense of part (c)). Show that it nevertheless gives the correct law by verifying a specific integral identity: for the test function $\varphi(x, y) = e^{-(x^2 + y^2)}$, show that

$$\int_{\mathbb{R}^2} \varphi d\mu_{(X,Y)} = \int_{\mathbb{R}} \varphi(t, \mu_Y + \frac{\sigma_Y}{\sigma_X}(t - \mu_X)) g(t) dt,$$

and explain why identities of this form, for a sufficient class of test functions, *determine the law* — that is, why any Borel probability measure μ' on $\mathbb{R}^2$ such that $\int \varphi d\mu' = \int \varphi d\mu_{(X,Y)}$ for every bounded continuous φ must equal $\mu_{(X,Y)}$ (you may quote that bounded continuous functions form a determining class for Borel probability measures on $\mathbb{R}^2$).

- (iii) Show that the conditional formulas of part (b) remain meaningful as $\rho^2 \rightarrow 1$: evaluate $\lim_{\rho \rightarrow 1} \mathbb{E}[X | Y = y]$ and $\lim_{\rho \rightarrow 1} \text{Var}[X | Y = y]$, and explain why the conditional description of part (b) survives even though the joint density does not.

Q2. [★] Absolute continuity of probability laws

Setup. A probability measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is *absolutely continuous with respect to a σ -finite reference measure ν* , written $\mu \ll \nu$, if

$$\nu(B) = 0 \implies \mu(B) = 0 \quad \text{for all } B \in \mathcal{B}(\mathbb{R}).$$

By the Radon–Nikodym theorem, $\mu \ll \nu$ iff there exists a nonnegative measurable $g = d\mu/d\nu$ (the *R–N derivative*) with $\mu(B) = \int_B g d\nu$ for all $B \in \mathcal{B}(\mathbb{R})$; such g is unique ν -a.e. Recall the worked method from Lecture 2 (“work from the defining identity”): to find g , test $B = \{a\}$ to read off atoms, then test intervals in the diffuse part. Also recall the bridge identity: if $\mu_X \ll \lambda$ and the cdf F_X is differentiable at x , then $F'_X(x) = d\mu_X/d\lambda(x)$ at that point. Throughout, $Z \sim \mathcal{N}(0, 1)$, λ is Lebesgue measure, and δ_a is the unit point mass at a .

- (a) For each of the following three laws μ_X of X , decide whether $\mu_X \ll \lambda$ and *prove* your answer from the definition (for the “yes” case, also produce g and verify the defining identity; for the “no” case, exhibit a witnessing λ -null set):

- (i) $X = e^Z$ (the log-normal law);
- (ii) $X = \max(Z, 0)$ (a rectified Gaussian sensor that saturates at 0);
- (iii) $X = \lfloor Z \rfloor$ (a uniform quantizer with unit step).

- (b) The two laws in (a)(ii)–(iii) are *not* absolutely continuous w.r.t. λ , but each becomes absolutely continuous after we *redefine the reference measure*.
- (i) For $X = \max(Z, 0)$, take $\nu := \lambda + \delta_0$. Verify $\mu_X \ll \nu$, and find $g = d\mu_X/d\nu$ from the defining identity by testing (1) the atom $B = \{0\}$; (2) intervals $B \subset (0, \infty)$. State the value of $g(0)$ and identify g on the rest of $\mathbb{R}$ (up to λ -a.e. equality).
 - (ii) For $X = \lfloor Z \rfloor$, take $\nu := \lambda + \kappa$ where $\kappa := \sum_{k \in \mathbb{Z}} \delta_k$ is counting measure on the integers. Verify $\mu_X \ll \nu$ and find $g = d\mu_X/d\nu$. (Pay attention to what g must do on the non-integer points, and why.) Check your g on the test set $B = (0.5, 2.5]$ by evaluating both sides of $\mu_X(B) = \int_B g d\nu$.
- (c) **Not-differentiable yet absolutely continuous.** Let $X = \min(Z, 1)$ (an upper-saturated Gaussian sensor). Let F_X be its cdf.
- (i) Show that $\mu_X \not\ll \lambda$, and that F_X is *not differentiable at $x = 1$* (compute the left derivative and the jump). Show also that F_X is differentiable at every $x \neq 1$.
 - (ii) With the modified reference $\nu = \lambda + \delta_1$, verify $\mu_X \ll \nu$ and find $g = d\mu_X/d\nu$. Show that the R–N derivative recovers the atom as a finite *point value*, $g(1) = 1 - \Phi(1) \approx 0.1587$, even though the classical derivative $F'_X(1)$ does not exist. Verify the defining identity on $B = [1, 2]$.
 - (iii) Explain in one or two sentences the resolution of the paradox: the classical derivative differentiates *functions* (and fails at jumps), while the R–N derivative differentiates *measures with respect to a chosen reference* (and succeeds as soon as the reference “charges” every set that μ_X charges).
- (d) **The no-atom theorem for Lebesgue-absolutely-continuous laws.**
- (i) Suppose $\mu_X \ll \lambda$. Prove that $\mathbb{P}(X = x) = 0$ for every $x \in \mathbb{R}$, and deduce $\mathbb{P}(X \leq x) = \mathbb{P}(X < x)$ for every $x \in \mathbb{R}$.
 - (ii) Extend: prove $\mathbb{P}(X \in C) = 0$ for every countable $C \subset \mathbb{R}$, and conclude that the cdf F_X is continuous everywhere. (Hint: countable additivity.)
 - (iii) **Sharpness.** Show by a counterexample that the converse implication is false: find a law with $\mathbb{P}(X = x) = 0$ for all x , yet $\mu_X \not\ll \lambda$. (You may take as given that the Cantor distribution — the law whose cdf is the Cantor function — exists and satisfies $\mathbb{P}(X = x) = 0$ for all x ; explain why $\mu_X \ll \lambda$ fails for it.)
 - (iv) Contrast with part (c): for $X = \min(Z, 1)$, evaluate $\mathbb{P}(X \leq 1) - \mathbb{P}(X < 1)$ and explain which hypothesis of (i) fails.

Q3. From Markov to Chernoff: tightening a tail bound, and reading off the exact regime

Setup. Let $X_1, X_2, \dots \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(\lambda)$ with density $f(x) = \lambda e^{-\lambda x} \mathbf{1}_{x \geq 0}$ and mgf $M(\theta) = \mathbb{E} e^{\theta X_1} = \frac{\lambda}{\lambda - \theta}$ for $\theta < \lambda$. Let $S_n = X_1 + \dots + X_n$; recall $\mathbb{E} S_n = \frac{n}{\lambda}$, $\text{Var} S_n = \frac{n}{\lambda^2}$, and that S_n has the Gamma density $f_{S_n}(x) = \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!} \mathbf{1}_{x \geq 0}$ (so that $S_n \sim \text{Gamma}(n, \lambda)$). Recall the toolkit of Lecture 2 §3: Markov ($\mathbb{P}(X \geq a) \leq \mathbb{E} X/a$ for $X \geq 0$), Chebyshev ($\mathbb{P}(|X - \mathbb{E} X| \geq t) \leq \text{Var} X/t^2$), Paley–Zygmund ($\mathbb{P}(X \geq \theta \mathbb{E} X) \geq (1 - \theta)^2 \frac{(\mathbb{E} X)^2}{\mathbb{E} X^2}$ for $X \geq 0$, $\theta \in [0, 1]$), and the Chernoff method ($\mathbb{P}(X \geq a) \leq \inf_{\theta > 0} e^{-\theta a} \mathbb{E} e^{\theta X}$). The running objective is a *non-asymptotic upper bound on the upper tail* $\mathbb{P}(S_n \geq a)$ for $a > n/\lambda$.

- (a) Compute the Markov bound $\mathbb{P}(S_n \geq a) \leq n/(\lambda a)$. Evaluate it at $a = cn/\lambda$ ($c > 1$) and at $a = (n + s\sqrt{n})/\lambda$ ($s > 0$). Show — with a one-line reason — that this bound can never fall below a positive constant as $n \rightarrow \infty$, whatever the deviation.
- (b) Apply Chebyshev’s inequality (hint: $\mathbb{P}(|X - \mathbb{E} X| \geq t) \leq \text{Var} X/t^2$) to obtain $\mathbb{P}(S_n \geq a) \leq \frac{n}{(\lambda a - n)^2}$ for $a > n/\lambda$. Evaluate it at $a = cn/\lambda$ and at $a = (n + s\sqrt{n})/\lambda$, and conclude that Chebyshev decays

only *polynomially* in n at the first scale and *not at all* at the second. Can the Paley–Zygmund inequality be used to produce a matching *lower* bound for the event $\{S_n \geq a\}$ here? Explain precisely why or why not (which sign of deviation does P–Z control?).

- (c) **Chernoff.** Derive the bound $\mathbb{P}(S_n \geq a) \leq B_n(a) := e^{-\theta a} \left(\frac{\lambda}{\lambda - \theta}\right)^n$ valid for $0 < \theta < \lambda$; optimise over θ to show the minimiser is $\theta^* = \lambda - \frac{n}{a}$ (verify $\theta^* \in (0, \lambda)$ when $a > n/\lambda$) and conclude

$$\mathbb{P}(S_n \geq a) \leq B_n(a) = \exp\left(-\left[\lambda a - n - n \ln \frac{\lambda a}{n}\right]\right) = \exp(-n I(\lambda a/n)), \quad I(x) := x - 1 - \ln x.$$

Show $I(x) > 0$ for $x > 0$, $x \neq 1$, and $I(1) = 0$ (convexity/minimum argument).

- (d) **Two scales of the deviation; which bound is useful where.**

- (i) **Linear scale** $a = cn/\lambda$, $c > 1$ fixed. Show that the exact tail satisfies $-\frac{1}{n} \ln \mathbb{P}(S_n \geq a) \rightarrow I(c)$, i.e. the Chernoff bound's exponent *is* the true large-deviation rate; contrast with (a)–(b) where the competitors' exponents vanish (constants; $-\ln(1/((c-1)^2 n)) = O(\ln n)$). Illustrate numerically for $\lambda = 1$, $c = 1.5$, $n = 100$: compute the exact tail (use $\mathbb{P}(S_n \geq a) = e^{-a} \sum_{k=0}^{n-1} a^k/k!$, which equals the Poisson CDF identity), Markov, Chebyshev and Chernoff bounds and compare all four numbers.

- (ii) **CLT scale** $a = (n + s\sqrt{n})/\lambda$, $s > 0$ fixed. Show by Taylor expansion that

$$-\ln B_n(a) = \frac{s^2}{2} - \frac{s^3}{3\sqrt{n}} + O\left(\frac{s^4}{n}\right).$$

Then use the central limit theorem to show that the exact tail satisfies $\mathbb{P}(S_n \geq a) \rightarrow \Phi(-s)$ as $n \rightarrow \infty$ at fixed s . Interpret: at the CLT scale the Chernoff bound matches the exact probability to exponential order with polynomial slack, while the Markov and Chebyshev bounds degrade to $\Theta(1)$ constants (evaluate them at this scale to confirm).

Q4. Mean-ergodicity of random processes: a case study

Setup. A real process $\{X_n\}_{n \geq 1}$ is *wide-sense stationary* (WSS) if $m := \mathbb{E}X_n$ is constant in n and $\text{Cov}(X_n, X_{n+k}) = \gamma_k$ depends only on k ; $\gamma_0 = \text{Var}X_n$. Define the running time average $\bar{X}_N := \frac{1}{N} \sum_{n=1}^N X_n$. The process is *mean-ergodic* (in L^2) if $\bar{X}_N \rightarrow m$ in L^2 , i.e. $\mathbb{E}|\bar{X}_N - m|^2 \rightarrow 0$. **Discrete Slutsky criterion (you may quote):** a WSS process is mean-ergodic iff

$$\text{Var}(\bar{X}_N) = \frac{1}{N^2} \sum_{i,j=1}^N \gamma_{i-j} = \frac{1}{N} \sum_{|k| \leq N-1} \left(1 - \frac{|k|}{N}\right) \gamma_k \xrightarrow{N \rightarrow \infty} 0,$$

and a sufficient condition is $\sum_{k \in \mathbb{Z}} |\gamma_k| < \infty$. Let $W_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_W^2)$ throughout, and let $\Theta \sim \mathcal{N}(\mu_\Theta, \sigma_\Theta^2)$ be independent of all W_n ($\mu_\Theta \in \mathbb{R}$ is the (unknown) ensemble mean level and $\sigma_\Theta^2 > 0$ the realization-to-realization spread; this is the standard “random DC” model).

- (a) **AR(1).** Let $X_n = \rho X_{n-1} + W_n$ with $|\rho| < 1$ (and the recursion initialised at the stationary law). Show $\mathbb{E}X_n = 0$, derive the recursion $\gamma_0 = \rho^2 \gamma_0 + \sigma_W^2$ and show that

$$\gamma_0 = \frac{\sigma_W^2}{1 - \rho^2}, \quad \gamma_k = \frac{\sigma_W^2}{1 - \rho^2} \rho^{|k|} \quad (k \in \mathbb{Z}).$$

Hence, prove that $\{X_n\}$ is WSS. (For γ_k , multiply the recursion by X_{n-k} and take expectations, or use induction on k .)

- (b) **Mean-ergodicity of AR(1).** Using the criterion, show that

$$\text{Var}(\bar{X}_N) = \frac{\gamma_0}{N^2} \left[N + 2 \sum_{k=1}^{N-1} (N - k) \rho^k \right],$$

and show that $\text{Var}(\bar{X}_N) \rightarrow 0$ as $N \rightarrow \infty$. Then show $\text{Var}(\bar{X}_N) \sim \frac{\sigma_W^2}{N(1-\rho)^2}$ as $N \rightarrow \infty$, check the formula at $\rho = 0$, and explain why the *effective sample size* — defined as $N_{\text{eff}} := \gamma_0 / \text{Var}(\bar{X}_N)$, i.e. the number of independent samples with variance γ_0 that would give the same estimator variance — satisfies $N_{\text{eff}} \approx N(1-\rho)^2$ rather than N (how far does the autocorrelation reach?).

- (c) **A non-ergodic process.** Let $X_n = \mu_\Theta + \Gamma + W_n$ where $\Gamma \sim \mathcal{N}(0, \sigma_\Theta^2)$: every realization has its own random DC level. Equivalently $X_n = \Theta + W_n$ with $\Theta := \mu_\Theta + \Gamma$. Show $\{X_n\}$ is WSS with mean $m = \mu_\Theta$ and $\gamma_k = \sigma_\Theta^2 + \sigma_W^2 \delta_{k0}$, and show via the criterion that it is *not* mean-ergodic: compute $\text{Var}(\bar{X}_N) = \sigma_\Theta^2 + \sigma_W^2/N$ and conclude $\mathbb{E}|\bar{X}_N - m|^2 \rightarrow \sigma_\Theta^2 \neq 0$. Identify explicitly a *shift-invariant event* whose probability is not 0 or 1 (Lecture 2, §5.6/Birkhoff), and explain why time-averaging can never remove Θ . What is the smallest possible value of $\lim_{N \rightarrow \infty} \mathbb{E}|\bar{X}_N - m|^2$ — the error floor that no amount of averaging (no choice of the window N) can beat — and explain the long-run behavior of $\bar{X}_N$ (what random variable does it converge to, and what does that imply about estimating μ_Θ from a single realization)?
- (d) **Differencing restores ergodicity — at a price.** Consider the differenced process $Z_n := X_n - X_{n-1}$ built from the process of part (c).

- (i) Show that $\{Z_n\}$ is WSS with $\mathbb{E}Z_n = 0$ and

$$\gamma_Z(0) = 2\sigma_W^2, \quad \gamma_Z(\pm 1) = -\sigma_W^2, \quad \gamma_Z(k) = 0 \quad (|k| \geq 2).$$

- (ii) Compute $\text{Var}(\bar{Z}_N)$ *exactly* for every N , and show $\text{Var}(\bar{Z}_N) = \frac{2\sigma_W^2}{N^2}$. Conclude that $\{Z_n\}$ is mean-ergodic, and note that its variance decays like N^{-2} — faster than any average of X could decay.
- (iii) Why does differencing annihilate the non-ergodic component, while plain averaging cannot? What does $\bar{Z}_N$ estimate, and what does it not? It is a consistent estimator of the mean of Z — which is 0 — but does it recover $m = \mathbb{E}X_n = \mu_\Theta$? Contrast with part (c): which is destroyed by the difference operator, the random level Θ itself or the possibility of estimating the ensemble level μ_Θ from one realization?
- (iv) Quantitative comparison. Take $\sigma_W = 1$ and $\sigma_\Theta = 0.1$. Find the smallest N^* such that $\text{Var}(\bar{Z}_{N^*}) \leq \inf_N \text{Var}(\bar{X}_N)$, i.e. the differenced average becomes strictly better than the best possible raw average, and evaluate N^* . Comment on the practical moral (“if a random DC component is suspected, ...”).

Q5. ISI channels I: block model, ZF, MMSE, and the CP-OFDM upgrade

Setup. A discrete-time intersymbol-interference (ISI) channel with L taps $h_0, \dots, h_{L-1}$ produces, for a transmitted symbol stream $\{x_n\}$,

$$y_n = \sum_{\ell=0}^{L-1} h_\ell x_{n-\ell} + w_n, \quad w_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, N_0),$$

with inputs $\mathbb{E}|x_n|^2 = E_s$, $x_k = 0$ for $k < 0$ or $k \geq N$ (finite block of length N ; a boundary convention you will use throughout). Run the numerical example

$$L = 4, \quad \mathbf{h} = (h_0, h_1, h_2, h_3) = (1, 0.65, 0.4, 0.2), \quad N = 6$$

whenever a numeric answer is asked. Recall from Lecture 3: *Toeplitz* banded structure, the block model $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{w}$; ZF zero-forcing means $\mathbf{W}\mathbf{H} = \mathbf{I}_N$ (exact ISI removal); MMSE minimises $\mathbb{E}\|\mathbf{W}\mathbf{y} - \mathbf{x}\|^2$; a *circulant* matrix $\mathbf{C}$ satisfies $[\mathbf{C}]_{ij} = c_{(i-j) \bmod N}$ and is diagonalised by the unitary DFT matrix $\mathbf{F}$, $[\mathbf{F}]_{k\ell} = \frac{1}{\sqrt{N}} e^{-j2\pi k\ell/N}$; the *cyclic prefix (CP)* of length $\mu \geq L - 1$ prepends a copy of the last μ symbols so that the block sees a circular convolution.

- (a) **Block model.** (i) Show that stacking $y_0, \dots, y_{N+L-2}$ gives $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{w}$ with $\mathbf{H}$ the $(N+L-1) \times N$ matrix $[\mathbf{H}]_{n,k} = h_{n-k}$ ($0 \leq n-k \leq L-1$, else 0); write $\mathbf{H}$ out *entry by entry* for the numerical example. (ii) Verify explicitly that $\mathbf{H}$ is Toeplitz (constant along diagonals) and band-limited (which diagonals are nonzero?). (iii) *Streaming penalty:* suppose the channel is used continuously with past symbols x_{-3}, x_{-2}, x_{-1} (the tail of the previous block) also present. Identify exactly which entries of the current block's output vector are polluted by inter-block interference (IBI), and write the correction term as a matrix-vector product involving $\mathbf{x}_{\text{prev}} = (x_{-3}, x_{-2}, x_{-1})^T$. How many output samples are affected, and with what weights (which channel taps multiply each past symbol in the correction term)?

- (b) **Zero forcing.** (i) Formulate the ZF equalizer as the optimisation problem

$$\min_{\mathbf{W} \in \mathbb{C}^{N \times (N+L-1)}} \|\mathbf{W}\|_F^2 \quad \text{subject to} \quad \mathbf{W}\mathbf{H} = \mathbf{I}_N,$$

and *solve* it: show $\mathbf{W}_{\text{ZF}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H$ is feasible, and then prove optimality by the perturbation argument: for any feasible $\tilde{\mathbf{W}} = \mathbf{W}_{\text{ZF}} + \Delta$ with $\Delta \mathbf{H} = \mathbf{0}$, expand $\|\tilde{\mathbf{W}}\|_F^2$ and show the cross term vanishes (use $\text{tr}(\mathbf{W}_{\text{ZF}} \Delta^H) = \text{tr}((\mathbf{H}^H \mathbf{H})^{-1} (\Delta \mathbf{H})^H)$). (ii) Compute $\mathbf{W}_{\text{ZF}}$ (numerically; show the matrix). (iii) The equalised noise $\mathbf{W}_{\text{ZF}} \mathbf{w}$ has covariance $N_0 (\mathbf{H}^H \mathbf{H})^{-1}$: compute the diagonal $[(\mathbf{H}^H \mathbf{H})^{-1}]_{kk}$ for $k = 1, \dots, 6$ and describe the *spatial profile* of noise enhancement along the block (which symbols suffer most? by what factor?). (iv) Explain why the profile would blow up as the smallest eigenvalue of $\mathbf{H}^H \mathbf{H}$ approaches zero, and relate this to deep notches of the frequency response $H(f) = \sum_{\ell} h_{\ell} e^{-j2\pi f \ell}$ of the channel.

- (c) **MMSE.** (i) Formulate the MMSE problem for i.i.d. inputs: $\min_{\mathbf{W}} \mathbb{E} \|\mathbf{W}\mathbf{y} - \mathbf{x}\|^2$ with $\mathbb{E}[\mathbf{x}\mathbf{x}^H] = E_s \mathbf{I}$, $\mathbb{E}[\mathbf{w}\mathbf{w}^H] = N_0 \mathbf{I}$, $\mathbb{E}[\mathbf{x}\mathbf{w}^H] = \mathbf{0}$. (ii) Apply the orthogonality principle (which you may quote: the optimal $\mathbf{W}$ satisfies $\mathbb{E}[(\mathbf{W}\mathbf{y} - \mathbf{x})\mathbf{y}^H] = \mathbf{0}$), and derive

$$\mathbf{W}_{\text{MMSE}} = \left(\mathbf{H}^H \mathbf{H} + \frac{N_0}{E_s} \mathbf{I}_N \right)^{-1} \mathbf{H}^H.$$

(iii) Prove the *dual form* $\mathbf{W}_{\text{MMSE}} = \mathbf{H}^H (\mathbf{H}^H \mathbf{H} + \frac{N_0}{E_s} \mathbf{I}_{N+L-1})^{-1}$ by direct matrix algebra (multiply through; this form shows one only inverts an $(N+L-1) \times (N+L-1)$ matrix). (iv) *Performance comparison.* Diagonalise $\mathbf{H}^H \mathbf{H} = \mathbf{V} \Lambda \mathbf{V}^H$ with $\lambda_1, \dots, \lambda_N > 0$. Show that the per-block mean-squared errors of the two equalizers are

$$\text{MSE}_{\text{ZF}} = N_0 \sum_{i=1}^N \frac{1}{\lambda_i}, \quad \text{MSE}_{\text{MMSE}} = N_0 \sum_{i=1}^N \frac{1}{\lambda_i + N_0/E_s},$$

and hence $\text{MSE}_{\text{MMSE}} \leq \text{MSE}_{\text{ZF}}$ with equality as $E_s/N_0 \rightarrow \infty$. Interpret: which eigendirections does MMSE repair, and why does the repair matter less and less at high SNR? Evaluate both at $E_s/N_0 = 10$ dB for the numerical example (in units of E_s), and compute the eigenvalues λ_i of $\mathbf{H}^H \mathbf{H}$ to display the per-direction comparison.

- (d) **CP-OFDM: from Toeplitz to circulant.** Keep $N = 6$, $\mathbf{h}$ as above, and take $\mu = L - 1 = 3$ if needed.

- (i) Write the 6×6 circulant $\mathbf{H}_{\text{circ}}$ whose first column is $(h_0, h_1, h_2, h_3, 0, 0)^T$ and verify that adding a CP of length $\mu = 3$ converts the block channel into $\mathbf{H}_{\text{circ}}$ (explain, sample by sample, how the CP samples “absorb” the inter-block tail; you may use the figure-style argument from the slides).
(ii) Show $\mathbf{F} \mathbf{H}_{\text{circ}} \mathbf{F}^H = \text{diag}(\Lambda_0, \dots, \Lambda_5)$ with $\Lambda_k = \sum_{\ell=0}^3 h_{\ell} e^{-j2\pi k \ell / 6}$, and compute the six complex numbers Λ_k and their magnitudes $|\Lambda_k|$.
(iii) Sketch the transmitter (symbol block $\rightarrow$ IFFT $\rightarrow$ CP insertion $\rightarrow$ P/S) and receiver (S/P $\rightarrow$ CP removal $\rightarrow$ FFT $\rightarrow$ per-bin equalisation $\hat{x}_k = \frac{1}{\Lambda_k} (\mathbf{F}\mathbf{y})_k \rightarrow$ decisions) block diagrams, and explain why the receiver “sees N independent flat channels”.

- (iv) *Analysis.* (A) Give a closed-form expression for the post-equalisation SNR of subcarrier k (*per-bin* ZF) and evaluate all six at $E_s/N_0 = 10$ dB. (B) Compute the aggregate spectral efficiency $\eta \cdot \frac{1}{N} \sum_{k=0}^{N-1} \log_2(1 + \frac{E_s}{N_0} |\Lambda_k|^2)$ where $\eta = N/(N + \mu)$ is the CP overhead factor; evaluate it (will be ≈ 2.4 bits/s/Hz). (C) Identify the weakest subcarrier and relate the *per-bin* ZF noise boost $1/|\Lambda_k|^2$ to the notch structure; state in two sentences how *per-bin* MMSE equalisation $\hat{x}_k = \frac{\Lambda_k^*}{|\Lambda_k|^2 + N_0/E_s} (\mathbf{F}\mathbf{y})_k$ repairs the weakest bins, and what this implies at very high SNR.

Q6. Dispersion parameters of a Jakes channel: delay spread, coherence bandwidth, Doppler spread, coherence time

Setup. For a WSSUS channel, the *power-delay profile* is $P(\tau) = \mathbb{E}|h(t, \tau)|^2$; the *mean excess delay* and *RMS delay spread* are

$$\bar{\tau} = \frac{\int_0^\infty \tau P(\tau) d\tau}{\int_0^\infty P(\tau) d\tau}, \quad \sigma_\tau = \sqrt{\overline{\tau^2} - \bar{\tau}^2}, \quad \overline{\tau^2} = \frac{\int_0^\infty \tau^2 P(\tau) d\tau}{\int_0^\infty P(\tau) d\tau}.$$

The *spaced-frequency correlation* and *coherence bandwidth* are $R_H(\Delta f) = \mathbb{E}[H^*(t, f)H(t, f + \Delta f)]$ with $H(t, f) = \int h(t, \tau) e^{-j2\pi f \tau} d\tau$, and B_c is the Δf at which $|R_H|$ first drops below a prescribed fraction (50%, or $1/e$) of $R_H(0)$. Dually, the *Doppler power spectrum* is $S(\nu) = \int R_h(\Delta t) e^{-j2\pi \nu \Delta t} d(\Delta t)$ where $R_h(\Delta t) = \mathbb{E}[h^*(t, \tau)h(t + \Delta t, \tau)]$, the *maximum Doppler shift* is $f_d = v f_c / c$, the *Doppler spread* B_d is the effective support width of $S(\nu)$, and the *coherence time* T_c is the Δt at which $|R_h(\Delta t)|$ first drops below a prescribed fraction of $R_h(0)$. Throughout, use the empirical rule $B_c \approx 1/(5\sigma_\tau)$ for the 50% level unless a problem asks for exact values.

- (a) **Delay spread from the PDP.** (i) For the 3-tap PDP $P(\tau) = \frac{1}{2}\delta(\tau) + \frac{1}{4}\delta(\tau - \Delta) + \frac{1}{4}\delta(\tau - 2\Delta)$, compute $\bar{\tau}$, $\overline{\tau^2}$ and σ_τ (in units of Δ ; verify $\sigma_\tau \approx 0.829\Delta$). (ii) For the continuous exponential PDP $P(\tau) = \frac{1}{\tau_0} e^{-\tau/\tau_0}$, show by explicit integration (integration by parts) that $\bar{\tau} = \tau_0$ and $\sigma_\tau = \tau_0$ — the exponential PDP has a single-parameter shape with mean equal to spread.
- (b) **Coherence bandwidth.** (i) Show $R_H(\Delta f) = \mathcal{F}\{P\}(\Delta f)$, the Fourier transform of the PDP (derive it from the WSSUS uncorrelated-scattering structure: $\mathbb{E}[h^*(t, \tau_1)h(t, \tau_2)] = P(\tau_1)\delta(\tau_1 - \tau_2)$). (ii) For the exponential PDP, compute $|R_H(\Delta f)| = (1 + (2\pi\Delta f\tau_0)^2)^{-1/2}$ explicitly, and find the exact $1/e$ and -3 dB crossing frequencies $0.4023/\tau_0$ and $1/(2\pi\tau_0)$. (iii) Compare with the empirical rule: show $1/(5\sigma_\tau)$ and $0.4023/\sigma_\tau$ differ by roughly a factor of 2, and state the relationship as “ $B_c\sigma_\tau$ is a constant of order 0.2–0.4, depending on the chosen coherence level”. Evaluate B_c for the 3-tap PDP of (a) with $\Delta = 0.4 \mu\text{s}$.
- (c) **Jakes Doppler.** (i) *Derive the U-shaped spectrum.* Assume isotropic scattering: rays arrive in azimuth with angle $\theta \sim \text{Unif}[0, 2\pi)$ and the Doppler shift of a ray is $\nu(\theta) = f_d \cos \theta$. Derive the density of ν by change of variables — handle the two-to-one nature of $\theta \mapsto \cos \theta$, e.g. via the standard result that $\cos \theta$ for uniform θ has density $\frac{1}{\pi\sqrt{1-c^2}} \mathbf{1}_{|c|<1}$ — and conclude $S(\nu) = \frac{1}{\pi f_d \sqrt{1-(\nu/f_d)^2}}$, $|\nu| < f_d$. Compute $\int_{-\infty}^\infty S(\nu) d\nu$ (verify it equals 1, so S is a density of unit rays), and note the divergence at $\nu = \pm f_d$ (why integrable?). (ii) *Two spreads.* Show that the second moment of the Jakes spectrum is $\int \nu^2 S(\nu) d\nu = \frac{1}{2} f_d^2$, so the *RMS Doppler spread* is $f_d/\sqrt{2}$ (substitute $\nu = f_d \sin \phi$). State the spread used by the course definition (support width: $B_d = 2f_d$) and comment on the difference between “support spread” and “RMS spread”. (iii) *Coherence time.* Derive $R_h(\Delta t) = J_0(2\pi f_d \Delta t)$ from $R_h(\Delta t) = \int S(\nu) e^{j2\pi \nu \Delta t} d\nu$ by the substitution $\nu = f_d \sin \theta$ (recognize the integral representation of the Bessel function $J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \sin \theta) d\theta$). Using the small-argument expansion $J_0(x) \approx 1 - x^2/4$, estimate T_c at the 50% level and compare with the exact value $0.242/f_d$ and with the first-zero benchmark $0.383/f_d$.
- (d) **Design synthesis.** A vehicle travels at $v = 90$ km/h on a carrier $f_c = 2$ GHz (take $c = 3 \times 10^8$ m/s). The propagation PDP is the 3-tap profile of (a) with $\Delta = 0.4 \mu\text{s}$. The system transmits with bandwidth $W_{\text{tx}} = 20$ MHz using frames of duration $T_{\text{frame}} = 1$ ms.

- (i) Compute the maximum Doppler shift f_d , the coherence times $T_c^{(0)} = 0.383/f_d$ and $T_c(50\%) = 0.242/f_d$, and the RMS delay spread σ_τ and coherence bandwidth $B_c \approx 1/(5\sigma_\tau)$.
- (ii) Compute the number of resolvable taps $L = \lceil \sigma_\tau W_{\text{tx}} \rceil$ and the CP length needed for a CP-OFDM air interface ($\mu \geq L - 1$).
- (iii) Place this channel on the four-regime map of Lecture 3 (axes: B_{tx} vs B_c ; T_{frame} vs T_c) and *justify* the regime. Then state the consequences: does the receiver need equalisation? time-tracking? both? none?
- (iv) If the same link were used with 10 ms frames (instead of 1 ms), would the regime classification change? Answer quantitatively, and explain the practical meaning of the boundary you are crossing. What does the gap between $B_d = 2f_d$ and a 15 kHz subcarrier spacing say about inter-carrier interference (ICI) for the design of (ii)–(iii)?

Q7. Log-normal shadowing, why exponents exceed 2, and cellular coverage

Setup. Large-scale propagation is modeled by

$$P_r(d)_{\text{dB}} = P_{t\text{dB}} + G_{\text{dB}} - \text{PL}_{\text{dB}}(d) - X_{\text{SH}}(d), \quad \text{PL}_{\text{dB}}(d) = \text{PL}_{\text{dB}}(d_0) + 10n \log_{10}(d/d_0),$$

where $\text{PL}_{\text{dB}}(d_0) = 20 \log_{10}(\frac{4\pi d_0}{\lambda})$ is the Friis free-space loss at the reference distance d_0 , n is the path-loss exponent, and $X_{\text{SH}} \sim \mathcal{N}(0, \sigma^2)$ is the (zero-mean, dB-scale) shadowing. Recall from Lecture 3: the shadowing arises as a sum of many independent per-obstacle attenuations $S = \sum_{i=1}^M \xi_i$ (each $\xi_i \geq 0$ in dB), whose CLT limit is $\mathcal{N}(m_M, s_M^2)$; the mean m_M is *merged into* the path-loss term, leaving the zero-mean fluctuation $X_{\text{SH}} = S - m_M$. Use the scenario

$$f_c = 1.8 \text{ GHz}, \quad d_0 = 100 \text{ m}, \quad n = 3.2, \quad \sigma = 8 \text{ dB}, \quad \text{EIRP} = P_t + G = 38 \text{ dBm}, \quad P_{\text{min}} = -85 \text{ dBm},$$

with P_{min} the receiver sensitivity. ($\lambda = cf_c^{-1}$; $c = 3 \times 10^8$ m/s.)

- (a) **Log-normal shadowing.** (i) Reproduce the CLT step: state why $S = \sum_i \xi_i$ is approximately Gaussian in dB (assumptions: independence, finite variances, many obstacles, Lindeberg-type negligibility), and why that makes the *linear* gain $G_{\text{SH}} = 10^{-X_{\text{SH}}/10}$ log-normal. (ii) Derive the log-normal density from the Gaussian: show that if $X \sim \mathcal{N}(0, \sigma^2)$ then $G = 10^{-X/10}$ has density

$$f_G(g) = \frac{10}{\ln 10} \cdot \frac{1}{g\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(10 \log_{10} g)^2}{2\sigma^2}\right), \quad g > 0.$$

- (iii) Compute $\mathbb{E}[G_{\text{SH}}]$ (use the Gaussian mgf $\mathbb{E}[e^{tX}] = e^{t^2\sigma^2/2}$): show the *mean* linear gain in dB exceeds the *median* (which is 0 dB) by $\frac{10}{\ln 10} \frac{\sigma^2(\ln 10)^2}{200} = \frac{\sigma^2 \ln 10}{20}$ dB, and evaluate for $\sigma = 8$ dB (you should find ≈ 7.4 dB). Explain why coverage predictions must use the *distribution* of X_{SH} rather than the mean path loss.
- (b) **Why terrestrial exponents exceed 2.** The measured exponent n in terrestrial links typically lies in $[2.7, 4]$, while free space gives $n = 2$.
 - (i) *Two-ray windowing.* Suppose that for $d > d_b$ (breakpoint) the loss obeys $n_2 = 4$ (ground reflection, $P_r \propto d^{-4}$), while for $d \leq d_b$ it is free-space. Show that fitting a *single-slope* model over $[d_0, D]$ with $D > d_b$ — defined mathematically as choosing one exponent n_{eff} so that the single-slope law reproduces the total dB loss between d_0 and D , i.e. $10n_{\text{eff}} \log_{10}(D/d_0) := \Delta \text{PL}_{\text{dB}}(d_0 \rightarrow D)$ — yields an apparent exponent

$$n_{\text{eff}} = \frac{20 \log_{10}(d_b/d_0) + 40 \log_{10}(D/d_b)}{10 \log_{10}(D/d_0)} = 2 + 2 \frac{\log_{10}(D/d_b)}{\log_{10}(D/d_0)} \in (2, 4),$$

and evaluate for $d_0 = 100$ m, $d_b = 300$ m, $D = 3$ km (you should find $n_{\text{eff}} \approx 3.35$).

- (ii) *Merging the shadowing mean.* Now let the shadowing have a distance-dependent dB mean, $\mathbb{E}[S(d)] = m(d)$, which the standard model merges into the path loss: $\text{PL}_{\text{eff}}(d) = \text{PL}(d) + m(d)$. Show that if $m(d) = \kappa \log_{10}(d/d_0)$ (κ dB per decade), then the merged model has constant apparent exponent $n_{\text{eff}} = n + \kappa/10$; and if instead $m(d) = \mu_* d$ (e.g. a mean number λd of blocking walls, each costing μ dB on average, so $\mu_* = \lambda\mu$), then the apparent exponent measured over one decade $[d_0, 10d_0]$ is $n_{\text{eff}} = n + \frac{9}{10}\mu_* d_0$, and in general $n_{\text{eff}}(d) = n + \frac{\mu_*(d-d_0)}{10 \log_{10}(d/d_0)}$ is *distance-dependent*. Evaluate both corrections for $\kappa = 8$ dB/decade, and for $\mu_* = 0.01$ dB/m with $d_0 = 100$ m; explain which model is consistent with the observed $n \in [2.7, 3.5]$ for macro cells, and why.
- (c) **Coverage of a base station.** Recall the coverage probability at distance d is $P_{\text{cov}}(d) = \mathbb{P}(P_r(d)_{\text{dB}} \geq P_{\text{min}}) = \Phi(\ell(d)/\sigma)$ where $\ell(d) = P_{\text{tdB}} + G_{\text{dB}} - \text{PL}_{\text{dB}}(d) - P_{\text{min}}$ is the margin against the *median* path loss (derive this from the setup).
- (i) Compute $\text{PL}_{\text{dB}}(d_0)$ (you should find ≈ 77.5 dB) and the margin $\ell(d_0)$ at the reference distance.
 - (ii) Solve for the cell radius R at which the *edge* coverage is 90% and 95%: use $\ell(R) = \sigma\Phi^{-1}(q)$, and report both radii (you should find ≈ 1.26 km and ≈ 1.02 km).
 - (iii) Quantify the reliability–range tradeoff: how many dB of margin does the jump $90\% \rightarrow 95\%$ cost, and by what factor does the radius shrink? (Give both numbers.)
- (d) **Area coverage and the marginal cost of reliability.** Users are uniformly distributed in the disk of radius R around the BS, and PL_{dB} is isotropic.
- (i) Derive the *area coverage*

$$P_{\text{area}}(R) = \frac{2}{R^2} \int_0^R r \Phi\left(\frac{\ell(r)}{\sigma}\right) dr,$$
explaining where the factor r (and the $2/R^2$) comes from.
 - (ii) Evaluate P_{area} at the 90%-edge radius of (c)(ii) by numerical quadrature and report it (you should find $\approx 96.4\%$). Explain why area coverage exceeds edge coverage.
 - (iii) *Closed form.* With the substitution $r = d_0 e^u$ and $B := \frac{10n}{\sigma \ln 10}$, show $P_{\text{area}}(R) = \Phi(\zeta) + e^{2\zeta/B+2/B^2} \Phi(-\zeta-2/B)$, where $\zeta := \ell(R)/\sigma$. (The key integral is $\int e^{2u} \Phi(a-Bu) du$; integrate by parts once and complete the square in the remaining Gaussian integral.) Verify your closed form reproduces the quadrature of (ii) at $R = 1.26$ km to 3 decimals.
 - (iv) *Marginal cost.* Keeping the same radius R as in (ii), how much extra EIRP (in dB) is needed to raise the edge coverage at that radius from 90% to 95%, and what does the area coverage become then? Comment on which is the more economical statement of a coverage target (a % at the edge, or a % over the area), citing both numbers you computed.

Q8. Diversity combining: SC, EGC, MRC and their asymptotics

Setup. A flat slow-fading link is equipped with L receive branches:

$$y_\ell = h_\ell x + w_\ell, \quad \ell = 1, \dots, L, \quad h_\ell \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, 1), \quad w_\ell \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, N_0), \quad x \in \mathbb{C}, \quad \mathbb{E}|x|^2 = E_s,$$

and a *linear combiner* forms $\hat{x} = \sum_{\ell=1}^L w_\ell^* y_\ell$ using the perfect knowledge of $\mathbf{h} = (h_1, \dots, h_L)$ at the receiver. Define the *outage probability* at threshold $\gamma_{\text{th}} > 0$ as $P_{\text{out}} = \mathbb{P}(\gamma < \gamma_{\text{th}})$, where γ is the SNR of the received statistic ($P_{\text{out}} > 0$ matters for a target rate R via the rate–outage relation, $\gamma_{\text{th}} = 2^{R-1}$); the *diversity order* of a scheme is $d := \lim_{\bar{\rho} \rightarrow \infty} \frac{-\ln P_{\text{out}}}{\ln \bar{\rho}}$. Denote the per-branch average SNR by $\bar{\rho} = E_s/N_0$, write the ℓ -th branch SNR $\gamma_\ell = |h_\ell|^2 \bar{\rho}$, and recall from Lecture 3: for $|h_\ell|^2 \sim \text{Exp}(1)$ (verify this from the Gaussianity of h_ℓ), $\gamma_\ell \sim \text{Exp}$ with mean $\bar{\rho}$; and the three classical combiners: selection (SC: pick $\max_\ell \gamma_\ell$), equal-gain (EGC: co-phase and add), and maximal-ratio (MRC).

- (a) **One branch.** (i) Show $\gamma_\ell \sim \text{Exp}(\bar{\rho})$ (density $f_\gamma(g) = \frac{1}{\bar{\rho}} e^{-g/\bar{\rho}}$; you may use $\mathbb{P}(|h|^2 > g) = e^{-g}$, itself

derivable from the Rayleigh survival function). (ii) Compute the single-branch outage $P_{\text{out}}^{(1)}(\gamma_{\text{th}}) = 1 - e^{-\gamma_{\text{th}}/\bar{\rho}}$ and its high-SNR approximation $\approx \gamma_{\text{th}}/\bar{\rho}$. (iii) Using the definition, show the diversity order of one branch is $d = 1$.

- (b) **MRC.** (i) *Optimality by Cauchy-Schwarz.* For fixed fading $\mathbf{h}$, the output SNR of a general linear combiner is $\gamma = \frac{|\sum_{\ell} w_{\ell}^* h_{\ell}|^2}{\sum_{\ell} |w_{\ell}|^2} \bar{\rho}$ (derive this from the combiner model). Prove $|\sum_{\ell} w_{\ell}^* h_{\ell}|^2 \leq (\sum_{\ell} |w_{\ell}|^2)(\sum_{\ell} |h_{\ell}|^2)$, identify the equality condition $\mathbf{w} \propto \mathbf{h}$, and conclude that MRC ($w_{\ell} = h_{\ell}$) achieves $\gamma_{\text{MRC}} = \sum_{\ell=1}^L \gamma_{\ell}$ — the maximum possible. (ii) *Distribution.* Show that $\gamma_{\text{MRC}} \sim \text{Gamma}(L, \bar{\rho})$ i.e.

$$f_{\gamma_{\text{MRC}}}(g) = \frac{g^{L-1}}{(L-1)! \bar{\rho}^L} e^{-g/\bar{\rho}}, \quad g \geq 0$$

(use the mgf: show $\mathbb{E}[e^{s\gamma_{\ell}}] = (1 - \bar{\rho}s)^{-1}$ and hence $\mathbb{E}[e^{s\gamma_{\text{MRC}}}] = (1 - \bar{\rho}s)^{-L}$, the Gamma mgf). (iii) *Outage.* Integrating the density from (ii) by parts, show

$$P_{\text{out}}^{\text{MRC}}(\gamma_{\text{th}}) = 1 - e^{-x} \sum_{k=0}^{L-1} \frac{x^k}{k!}, \quad x := \frac{\gamma_{\text{th}}}{\bar{\rho}}.$$

- (c) **SC and friends: comparisons.** (i) Show $F_{\gamma_{\text{SC}}}(g) = (1 - e^{-g/\bar{\rho}})^L$ and hence $P_{\text{out}}^{\text{SC}}(\gamma_{\text{th}}) = (1 - e^{-x})^L$. (ii) Prove the *realization-wise* (same fading $\mathbf{h}$) dominance $\gamma_{\text{SC}} \leq \gamma_{\text{MRC}}$ and conclude $P_{\text{out}}^{\text{MRC}} \leq P_{\text{out}}^{\text{SC}}$ for every γ_{th} and every $\bar{\rho}$. (One inequality, one line — find it.) (iii) For EGC, $\gamma_{\text{EGC}} = \frac{(\sum_{\ell} |h_{\ell}|^2)}{L} \bar{\rho}$. Show $\gamma_{\text{EGC}} \leq \gamma_{\text{MRC}}$ realization-wise via Cauchy-Schwarz (when is equality achieved?), and then show *this argument does not extend to SC*: find an explicit realization of $\mathbf{h}$ (e.g. one dominant branch and $L-1$ tiny ones) for which $\gamma_{\text{EGC}} < \gamma_{\text{SC}}$. Interpret: combining with equal gains can lose to discarding branches; why does this not contradict “EGC achieves the same diversity order as SC”?

- (d) **High-SNR asymptotics: the price of selection, and the gain of coherence.**

- (i) Show that as $x \rightarrow 0$ (high SNR),

$$P_{\text{out}}^{\text{MRC}} \sim \frac{x^L}{L!}, \quad P_{\text{out}}^{\text{SC}} \sim x^L,$$

by expanding the exact formulas of (b)(iii) and (c)(i). (For MRC use $1 - e^{-x} \sum_{k < L} x^k/k! = e^{-x} \sum_{k \geq L} x^k/k!$ and expand.)

- (ii) Compute the diversity orders of SC, EGC and MRC from the asymptotics and state what all three have in common (which single phenomenon sets the order?).
(iii) *The coherence gain.* At a fixed outage target P_{out} and high SNR, the extra SNR that MRC saves relative to SC is the solution of $\bar{\rho}_{\text{SC}}^L / \bar{\rho}_{\text{MRC}}^L \approx L!$; show the power gain in dB is

$$\Delta_{\text{MRC-SC}} = \frac{10 \log_{10}(L!)}{L} \text{ dB},$$

and tabulate it for $L = 2, 3, 4$ (you should find $\approx 1.5, 2.6, 3.5$ dB). Explain the growth pattern: why does the coherence gain grow with L but very slowly (what function of L is it)?

- (iv) *Numbers.* Take $\gamma_{\text{th}} = 0$ dB. Compute the exact outage probabilities of SC and MRC for $L = 2$ and $L = 4$ at $\bar{\rho} = 10$ dB and 20 dB, and compare against the asymptotic formulas of (i). Report your results as a small table and state at which SNR the asymptotics become accurate to within a few percent.